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FA22-BCS-153

Theory Assignment 01

Question no 1:Solve the questions given below using Bisection as well as False Position methods.

(i).

$$f(x) = cos(x) - xnear - 2.$$

Solution

$f(-2) = cos(-2) - -2$

$f(-2) = 2.99$

Trying values near −2

$f(-3) = cos(-3) - -3$

$f(-3) = 3.99$

$f(-1) = cos(-1) - -1$

$f(-1) = 1.99$

As the unction is positive around -2, -3, -1 So this cannot be solved Because either a and b need to have opposite signs

(ii) $ln(x^4) = 7$ near 0.5 and 2.

Solution:

$f(x) = 4ln(x) - 7$

$f(a) = f(0.5) = 4ln(0.5) - 7 = -9.7724 < 0$

$f(b) = f(2) = 4ln(2) - 7 = -4.2274 < 0$

As both f(a) and f(b) < 0, so this cannot be solved as the signs of f(a) and f(b) should be opposite.

(iii).

$$x^3 = 55$$

$f(x) = x^3 - 55$

Choosing interval : $3^3 = 27$

$4^4 = 64$

$27 < 55 < 64$

So root lies between 3 & 4

Let $a = 3$, $b = 4$

f(3)= $3^3 - 55 = -28 < 0$

f(4)= $4^4 - 55 = 9 > 0$

Applying Bisection Method

$m = \frac{a+b}{2}$

$m = \frac{3+4}{2}$

$m = \frac{7}{2} = 3.5$

$f(m) = f(3.5) = 3.5^3 - 55$

$f(m) = f(3.5) = 42.875 - 55 = -12.125 < 0$

$f(a) < 0$

$f(b) > 0$

$f(m) < 0$

Root lies in upper interval

(m,b)=(3.5,4)

Applying Falsi method

$m = \frac{af(b) - bf(a)}{f(b) - f(a)}$

$m = \frac{3(9)-4(-28)}{9-(-28)}$

$m = 3.7568$

$f(m) = f(3.7568) = -1.97 < 0$

$f(a) < 0$

$f(b) > 0$

$f(m) < 0$

Root lies in upper interval

(m,b)=(3.5,4)

Question no 2: Use the bisection method to find a root of the following equation between $x = 0 and x = 2$

$$f(x) = e^{-x/4}(2-x) - 1$$

Solution

Here $a = 0$, $b = 2$

$f(a) = f(0) = e^{-0/4}(2-0) - 1$

$f(a) = f(0) = 1$

$f(b) = f(2) = e^{-2/4}(2-2) - 1$

$f(b) = f(2) = -1$

Now through Bisection method

$m = \frac{a+b}{2}$

$m = \frac{0+2}{2}$

$m = \frac{2}{2} = 1$

$f(m) = f(1) = e^{-1/4}(2-1) - 1$

$f(m) = f(1) = -0.2212$

Hence the subintervals are $(a, m) = (0, 1)$ and $(m, b) = (1, 2)$

$f(a) > 0$

$f(b) < 0$

$f(m) < 0$

So the root lies in lower sub interval $(a, m) = (0, 1)$

Question no 4:Find the root of $x^3 = 5$ using Bisection method. Hint: Cube both sides to proceed.

$$f(x) = x^3 - 5 = 0$$

Solution

Here let $a = 0$, $b = 3$

$f(a) = f(0) = 0^3 - 5$

$f(a) = f(0) = -5$

$f(b) = f(2) = 3^3 - 5$

$f(b) = f(2) = 22$

As $a = -5$ and $b = 22$ have opposite signs means our guess is correct

Now through Bisection method

$m = \frac{a+b}{2}$

$m = \frac{0+3}{2}$

$m = \frac{3}{2} = 1.5$

$f(m) = f(1.5) = 1.5^3 - 5$

$f(m) = f(1.5) = -1.625$

Hence the subintervals are $(a, m) = (0, 1.5)$ and $(m, b) = (1.5, 3)$

$f(a) < 0$

$f(b) > 0$

$f(m) < 0$

So the root lies in upper sub interval $(m, b) = (1.5, 3)$

Question no 5:Find the root of $f(x) = -12 - 21x + 18x^2 + 2.75x^3$

(a) Graphically,

(b) Using bisection method, and

(c) Using false position method.

Choose the initial guesses of −1 and 0.

Solution

$f(-1) = -12 - 21(-1) + 18(-1)^2 + 2.75(-1)^3 = 24.25 > 0$

$f(0) = -12 - 21(0) + 18(0)^2 + 2.75(0)^3 = -12 < 0$

Now through Bisection method

$m = \frac{a+b}{2}$

$m = \frac{-1+0}{2}$

$m = \frac{-1}{2} = -0.5$

$f(m) = f(-0.5) = -12 - 21(0.5) + 18(0.5)^2 + 2.75(0.5)^3 = 2.6563 > 0$

$f(a) > 0$

$f(b) < 0$

$f(m) > 0$

So the root lies in upper sub interval $(m, b) = (-0.5, 0)$

Using Falsi Position Formula

Substitute values

$$m = \frac{(-1)(-12)-(0)(24.25)}{-12-24.25}$$

$$m = \frac{12-0}{-36.25}$$

$$m = -0.3310$$

Now compute $f(m)$

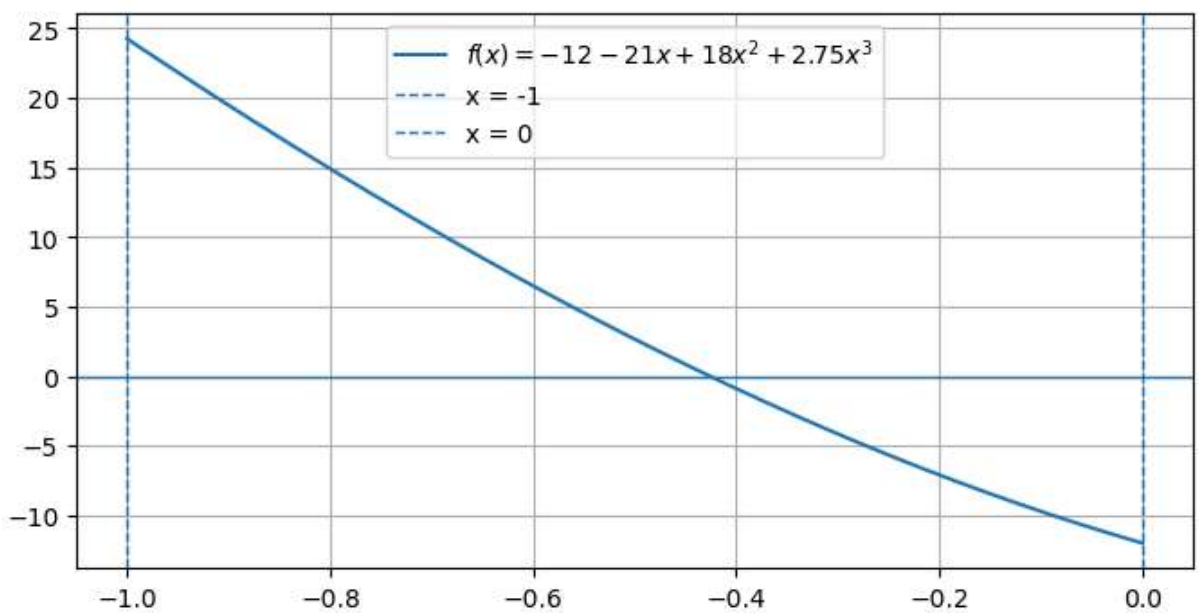
$$f(-0.3310) = -12 - 21(-0.3310) + 18(-0.3310)^2 + 2.75(-0.3310)^3$$

$$f(-0.3310) \approx -1.563$$

Since $f(m) < 0$ and $f(a) > 0$

The root lies in $(a, m) = (-1, -0.3310)$

In [2]:



Question no 3: Create a plot of f(x) = x^2|sin(x)| − 4 over the range 0 ≤ x ≤ 4. Then use the false position

method to determine a root that you observe in the plot.

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

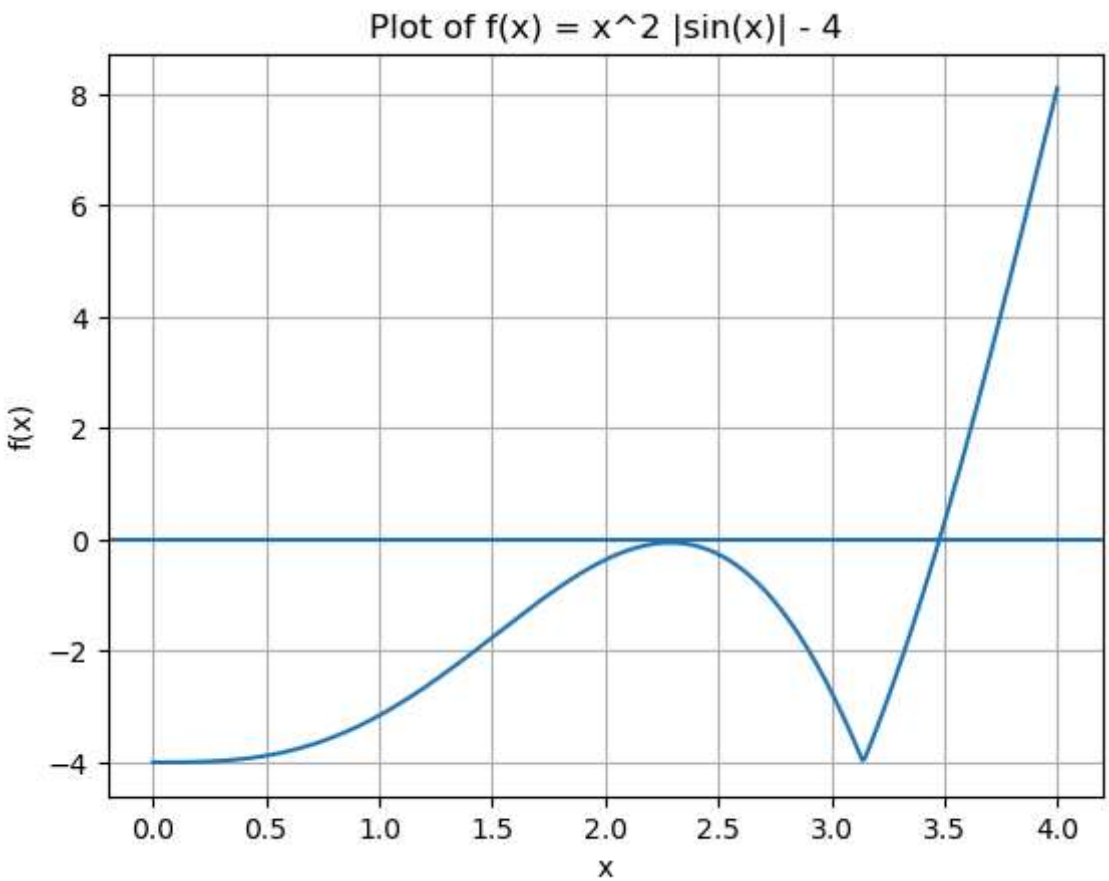
```python
# Define the function
def f(x):
    return x**2 * np.abs(np.sin(x)) - 4

# Create x values from 0 to 4
x = np.linspace(0, 4, 400)

# Compute y values
y = f(x)

# Plot the function
plt.figure()
plt.plot(x, y)
plt.axhline(0)
plt.title("Plot of f(x) = x^2 |sin(x)| - 4")
plt.xlabel("x")
plt.ylabel("f(x)")
plt.grid(True)

# Show the plot
plt.show()
```



Question no 3: Use the bisection method to find a root of the following equation between $x = 0 and x = 2$

$$\$f(x) = x^2|sin(x)| - 4$$

$

Solution

Here $a = 0$, $b = 4$

$f(a) = f(0) = 0^2|sin(0)| - 4$

$f(a) = f(0) = -4$

$f(b) = f(4) = 4^2|sin(4)| - 4$

$f(b) = f(2) = -2.8838$

As both are same sign negative sign so root does not exists